

AL AIN INTL, United Arab Emirates

WMO: 412180

Lat: 24.262N

Lon: 55.609E

Elev: 265

StdP: 98.18

Time Zone: 4.00 (E04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.0	12.1	-4.2	2.7	29.9	-2.0	3.3	30.5	11.0	22.8	9.5	21.3	2.7	40	0.495

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.2	46.0	22.5	45.1	22.6	44.1	22.6	28.8	35.8	28.0	35.6	27.1	35.6	4.4	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
27.2	23.8	32.3	26.1	22.2	32.1	25.0	20.8	32.2	96.3	35.6	91.7	35.4	87.6	35.6	32.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.3	9.0	8.1	8.7	47.5	1.2	0.6	7.9	48.0	7.2	48.3	6.6	48.7	5.8	49.1		
			5.6	30.0	1.6	1.1	4.5	30.8	3.6	31.5	2.7	32.1	1.6	32.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	29.2	18.8	21.0	24.4	29.3	33.9	36.3	37.5	37.4	34.7	30.7	25.1	20.8	(6)
(7)		DBStd	7.07	2.36	3.25	3.57	3.41	2.52	2.04	1.89	1.69	1.80	1.82	2.25	2.32	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	38	22	9	1	0	0	0	0	0	0	0	0	6	(9)
(10)		CDD10.0	7008	274	309	446	578	740	788	853	850	741	641	453	334	(10)
(11)		CDD18.3	4004	38	85	188	328	482	538	595	592	491	383	203	82	(11)
(12)		CDH23.3	60672	248	736	2054	4488	7729	9113	10344	10284	7944	5281	1901	549	(12)
(13)	CDH26.7	41408	22	235	988	2723	5417	6743	7867	7805	5568	3138	789	114	(13)	
(14)	Wind	WSAvg	3.9	3.5	4.0	4.3	4.3	4.2	4.1	4.2	4.0	3.7	3.4	3.4	(14)	
(15)	Precipitation	PrecAvg	73	11	9	18	8	2	1	5	4	3	1	2	8	(15)
(16)		PrecMax	169	70	39	74	60	3	11	60	12	24	7	29	65	(16)
(17)		PrecMin	10	0	1	0	0	1	0	0	2	0	0	0	0	(17)
(18)		PrecStd	46	18	9	24	13	1	3	12	2	5	2	6	17	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.7	33.2	37.8	41.5	45.7	47.0	47.0	46.2	44.6	40.8	35.1	31.0	(19)
(20)			MCWB	16.8	17.9	18.6	20.2	21.2	22.0	22.6	22.7	22.0	20.5	19.6	16.6	(20)
(21)		2%	DB	27.1	31.2	35.8	40.1	43.9	45.8	46.0	45.2	43.1	39.2	33.3	29.1	(21)
(22)			MCWB	16.6	17.4	18.2	19.6	21.2	21.9	23.1	22.9	21.9	20.1	18.9	17.2	(22)
(23)		5%	DB	26.0	29.7	34.0	38.8	42.8	44.8	45.0	44.7	42.1	38.1	32.3	28.0	(23)
(24)			MCWB	16.0	17.0	18.1	19.5	21.0	21.9	23.2	23.0	21.8	19.8	18.9	17.3	(24)
(25)		10%	DB	24.8	27.9	32.0	37.1	41.2	43.8	44.0	43.6	41.0	37.0	31.1	26.6	(25)
(26)			MCWB	15.6	16.6	17.2	19.0	20.7	21.9	23.2	23.2	21.8	19.9	18.8	17.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.7	21.2	22.6	24.1	25.9	28.7	30.1	29.8	28.7	26.3	23.7	21.1	(27)
(28)			MADB	24.5	27.2	30.3	34.4	35.9	36.0	37.0	35.9	33.8	30.0	28.4	25.8	(28)
(29)		2%	WB	18.7	19.6	21.2	22.6	24.6	27.2	29.0	28.6	27.4	25.2	22.4	19.9	(29)
(30)			MADB	23.8	26.5	28.6	33.5	35.5	35.3	36.4	35.6	33.0	29.8	27.5	24.6	(30)
(31)		5%	WB	17.8	18.7	20.2	21.5	23.7	26.3	28.1	27.7	26.5	24.3	21.6	19.1	(31)
(32)			MADB	23.2	26.0	27.7	32.7	35.5	35.1	36.2	36.0	32.8	29.8	27.0	24.5	(32)
(33)		10%	WB	16.9	17.8	19.3	20.6	22.8	25.4	27.2	26.7	25.6	23.4	20.9	18.4	(33)
(34)			MADB	22.2	24.9	27.6	32.1	35.7	35.2	36.3	36.5	33.0	29.9	26.8	23.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.7	11.7	12.7	13.5	14.7	14.8	13.2	12.7	13.1	12.6	11.3	10.9	(35)
(36)			MADB	11.7	12.8	14.1	14.3	15.7	15.5	13.7	13.4	13.8	13.5	12.2	12.1	(36)
(37)			MCWBR	4.9	5.2	5.3	5.5	5.5	5.6	5.4	5.3	5.8	6.3	4.9	4.8	(37)
(38)		5% WB	MADB	9.9	10.9	11.6	12.9	15.1	15.0	13.7	13.3	13.8	13.1	11.8	10.4	(38)
(39)			MCWBR	5.0	5.2	5.1	5.7	5.9	5.7	5.3	5.6	6.2	6.1	4.9	4.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.417	0.470	0.534	0.571	0.569	0.630	0.719	0.688	0.585	0.481	0.428	0.410	(40)	
(41)		taud	2.058	1.900	1.743	1.694	1.694	1.602	1.495	1.542	1.697	1.903	2.052	2.080	(41)	
(42)		Ebn at Noon	825	806	775	757	755	704	644	663	723	784	806	814	(42)	
(43)		Edh at Noon	149	187	229	246	245	267	296	282	237	183	148	140	(43)	
(44)	All-Sky Solar Radiation	RadAvg	4.36	5.14	6.02	6.77	7.51	7.59	7.02	6.87	6.46	5.70	4.67	4.14	(44)	
(45)		RadStd	0.23	0.32	0.36	0.39	0.17	0.19	0.25	0.16	0.12	0.10	0.10	0.21	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
Regional (12 neighbors)	+0.47	+0.70	N/A	N/A	N/A	N/A	N/A	N/A	+150	+149	

Nomenclature: See separate page